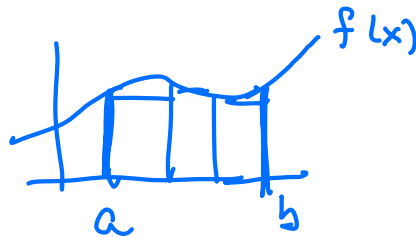


Class 36 The Riemann Integral

Goals

1. Know the basic idea of approximating a definite integral by a Riemann sum.
2. Understand terminology and notation for partitions, upper sums, and lower sums.
3. Know the definition of what it means for a function to be integrable on interval $[a, b]$.
4. Be able to use Darboux's Theorem to calculate a definite integral.

Basic Idea



Definite integrals arose in trying to calculate the area bounded by a curved line.

The strategy is to approximate the area under the curve by adding up areas of rectangles.

Area of rectangle = (base) · (height) where

the base is an interval on the x-axis and

the height is determined by the function value.

Dealing with bases

A partition of the interval $[a, b]$ is a finite set of numbers

$$P = \{x_0, x_1, \dots, x_n\} \text{ where } x_0 = a < x_1 < x_2 < \dots < x_n = b$$

If P and Q are two partitions of $[a, b]$, then Q is a refinement of P when the set P is a proper subset of Q .

Example: Give three distinct partitions P , Q , and R of the interval $[1, 7]$ where Q is a refinement of P with at least one irrational number and R is not a refinement of P

$$P =$$

$$Q =$$

$$R =$$

Dealing with heights: A real number M is an upper bound for a set of real numbers S if and only if

$$a \leq M \text{ for every } a \text{ in } S$$

If a set of real numbers S has an upper bound, then the supremum of S , denoted by **sup S** is the “least upper bound” satisfying two conditions:

$$1. a \leq \sup S \text{ for every } a \text{ in } S$$

$$2. \forall \epsilon > 0 \exists a \in S \text{ such that } a + \epsilon > \sup S$$

Give examples of sets of real numbers satisfying

a) A set that is not bounded above.

$$\mathbb{R}$$

b) An uncountable set, bounded above and containing its supremum of 4.

$$S = \{x \in \mathbb{R} : 0 \leq x \leq 4\} \quad [0, 4]$$

c) An uncountable set, bounded above but not containing its supremum of 5.

$$S = \{x \in \mathbb{R} : 0 \leq x < 5\} \quad [0, 5)$$

d) A countable set, bounded above but not containing its supremum of 6.

$$S = \{6 - \frac{1}{n} : n \in \mathbb{N}\}$$

e) A countable set, bounded above and containing its supremum of 7.

$$S = \{7 - \frac{1}{n} : n \in \mathbb{N}\} \cup \{7\}$$

Fill in similar definitions for lower bound and infimum for a set S .
Use lower case **m** to denote a lower bound for S and **inf S** to denote the infimum

Approximating integrals for a function f using partition $P = \{x_0, x_1, \dots, x_n\}$ for interval $[a, b]$.

Upper Riemann Sums

$$M_i(f) = \sup \{ f(x) : x \in [x_{i-1}, x_i] \} = \text{height of } i^{\text{th}} \text{ rectangle}$$

$$\Delta x_i = x_i - x_{i-1} = \text{width of } i^{\text{th}} \text{ rectangle}$$

$$\mathcal{U}(f, P) = \sum_{i=1}^n \overbrace{M_i(f) \Delta x_i}^{\text{Area of } i^{\text{th}} \text{ rectangle.}}$$

upper bound for value of the integral $\int_a^b f(x) dx$

Lower Riemann Sums

$$m_i(f) = \inf \{ f(x) : x \in [x_{i-1}, x_i] \} \quad \text{height}$$

$$\Delta x_i = x_i - x_{i-1} \quad \text{width}$$

$$\mathcal{L}(f, P) = \sum_{i=1}^n \overbrace{m_i(f) \Delta x_i}^{\text{area}}$$

Lower bound for $\int_a^b f(x) dx$

$$\mathcal{L}(f, P) \leq \int_a^b f(x) dx \leq \mathcal{U}(f, P)$$

Example: Approximate $\int_1^7 x+1 \, dx$ using $P = \{1, 3, 7\}$ and $Q = \{1, 3, 5, 7\}$



P	upper height	width	Area of upper rect	$U(f, P)$
$[1, 3]$	$M_1(f) = 4$	$\Delta x_1 = 2$	$4 \cdot 2 = 8$	$8 + 32 = 40$
$[3, 7]$	$M_2(f) = 8$	$\Delta x_2 = 4$	$8 \cdot 4 = 32$	

P	lower height	width	Area of lower rect	$L(f, P)$
$[1, 3]$	$m_1(f) = 2$	$\Delta x_1 = 2$	4	$4 + 16 = 20$
$[3, 7]$	$m_2(f) = 4$	$\Delta x_2 = 4$	16	

$$20 \leq \int_1^7 x+1 \, dx \leq 40$$

Q	$M_i(f)$	Δx_i	A_i	$U(f, Q)$
$[1, 3]$	4	2	8	$8 + 12 + 16 = 36$
$[3, 5]$	6	2	12	
$[5, 7]$	8	2	16	

Q	$m_i(f)$	Δx_i	A_i	$L(f, Q)$
$[1, 3]$	2	2	4	24
$[3, 5]$	4	2	8	
$[5, 7]$	6	2	12	

$$24 \leq \int_1^7 x+1 \, dx \leq 36$$